

# Protein Carbonylation Sites Identification by Classification Models – A Comparison

Natalia Rózga, Kaja Wróblewska

June 15, 2025

## Contents

|          |                                                        |           |
|----------|--------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                    | <b>2</b>  |
| <b>2</b> | <b>Data Overview and Preprocessing</b>                 | <b>2</b>  |
| 2.1      | Data Imbalance . . . . .                               | 2         |
| <b>3</b> | <b>Models and evaluations</b>                          | <b>3</b>  |
| 3.1      | Classical Supervised Learning Models . . . . .         | 3         |
| 3.2      | Deep Learning Model . . . . .                          | 3         |
| 3.3      | Evaluation metrics . . . . .                           | 4         |
| <b>4</b> | <b>Classical Supervised Learning Models - Analysis</b> | <b>4</b>  |
| 4.1      | Description . . . . .                                  | 4         |
| 4.2      | Results . . . . .                                      | 4         |
| <b>5</b> | <b>Deep learning models - LSTM</b>                     | <b>6</b>  |
| 5.1      | Description . . . . .                                  | 6         |
| 5.2      | Results . . . . .                                      | 7         |
| <b>6</b> | <b>Conclusion</b>                                      | <b>7</b>  |
| <b>7</b> | <b>Supplementary Data</b>                              | <b>9</b>  |
| 7.1      | Data indenpence . . . . .                              | 9         |
| 7.2      | Graphs for remaining amino acids . . . . .             | 10        |
| <b>8</b> | <b>Bibliography</b>                                    | <b>10</b> |

# 1 Introduction

Protein carbonylation is an irreversible oxidative modification linked to aging, neurodegenerative diseases, and cancer. The reliable identification of carbonylation sites is crucial for understanding the molecular mechanisms of oxidative stress. Given the vast amount of unprocessed data, machine learning approaches have become a powerful tool to classify peptide sequences based on their likelihood of undergoing carbonylation. This study compares various classification models for identifying carbonylation sites, placing special emphasis on **class balancing techniques**.

## 2 Data Overview and Preprocessing

The data used in this project comes from an "iCarPS" paper by Dan Zhang [1] and presents sequences categorized by carbonylation site predictions (K, R, T, P) labeled as either **positive** (carbonylated) or **negative** (non-carbonylated). The dataset is hierarchically organized by training/testing splits and amino acid categories. Key preprocessing steps included:

- **Data Melting:** Transforming the dataset from wide to long format and assigning binary class labels.
- **Sequence extraction:** "Sequences included unidentified amino acids denoted as *X*. To assess the impact of these missing residues, two strategies were applied: (1) omitting the *X* characters and analyzing the remaining sequence, and (2) removing the entire sequence record.

### 2.1 Data Imbalance

The dataset was significantly imbalanced, with a strong bias toward the *Negative* class. Since sequence-based data are not suitable for traditional resampling methods like *SMOTE*, balancing was addressed at the classifier level. Additionally, baseline references were introduced (*Dummy Classifier*) to better contextualize model performance.

| Group    | Dataset  | Carbonylation sites |     |     |     |
|----------|----------|---------------------|-----|-----|-----|
|          |          | K                   | P   | R   | T   |
| Training | Positive | 266                 | 114 | 119 | 116 |
|          | Negative | 1802                | 716 | 754 | 702 |
| Testing  | Positive | 34                  | 12  | 17  | 5   |
|          | Negative | 147                 | 76  | 93  | 30  |

Figure 1: Data distribution

## 3 Models and evaluations

### 3.1 Classical Supervised Learning Models

- Gaussian Naive Bayes (*Gaussian NB*)
- Random Forest
- Logistic Regression
- Support Vector Machine (*SVM*)

To establish a baseline and address the issue of imbalanced data, the **DummyClassifier** was implemented with the following properties:

- *most\_frequent* – always predicts the most frequent class observed in the training data. The accuracy ranged between 82 – 86%.
- *uniform* – randomly predicts a class with equal probability, regardless of class distribution. This approach achieved roughly 35 – 49% accuracy.

#### Feature extraction

##### Features based on sequence

Protein sequence features were extracted using Biopython’s *ProteinAnalysis* tool. This included biochemical properties such as molecular weight, isoelectric point, instability index, hydrophobicity, aromaticity, and secondary structure propensities (helix, turn, and sheet fractions).

##### Features based on amino acids

Additional amino acid features were derived from the *AAindex* database — a collection of numerical indices representing various physicochemical and biochemical properties of amino acids, based on experimental data from the literature. These features included properties such as hydrophobicity, side chain volume, molecular weight, charge, polarity, solvent accessibility, and flexibility. For each amino acid in a sequence, a vector representing the selected properties was extracted. The final representation of a sequence was then computed as either the mean or the sum of its amino acid vectors.

### 3.2 Deep Learning Model

A simple Long Short-Term Memory (LSTM) neural network was used as the deep learning model for sequence classification.

#### Feature extraction

Features were based on **Features based on amino acids** from Classical Supervised Learning Methods and left as embeddings.

### 3.3 Evaluation metrics

Evaluation was performed using two complementary metrics. First, the accuracy score was calculated with the DummyClassifier acting as a baseline reference. Second, the default F1-score (*average = 'binary'*) was used to account for class imbalance and provide a more detailed assessment of model performance. This yields the prediction score for the positive label - the unrepresented category.

## 4 Classical Supervised Learning Models - Analysis

### 4.1 Description

#### Gaussian Naive Bayes

GaussianNB assumes feature independence (check for independence in the **Supplementary Data [7.1]**) and Gaussian-distributed features within each class, allowing for fast and efficient training. However, the model does not support class balancing.

#### Random Forest

Random Forest is an ensemble method that builds multiple decision trees and averages their predictions, often leading to strong performance and susceptibility to overfitting. With *class\_weight = 'balanced'*, it assigns higher weights to minority class samples during tree construction, improving its sensitivity to rare classes. However, the majority voting mechanism may still bias the final prediction toward the majority class, especially when the imbalance is severe.

#### Logistic Regression

Logistic regression estimates class probabilities through a logistic function. The *class\_weight = 'balanced'* was used, allowing the algorithm to adjust weights inversely proportional to class frequencies, making it more resistant to class imbalance; however, its linear decision boundary may limit performance on more complex data distributions.

#### Support Vector Machine

The SVM classifier searches for the optimal hyperplane that maximizes the margin between classes, often yielding strong performance in high-dimensional spaces. *class\_weight = 'balanced'* was applied, which automatically adjusts weights inversely proportional to class frequencies in the input data.

### 4.2 Results

The Gaussian Naive Bayes classifier yielded results close to those of the dummy classifier, with an F1-score near zero, indicating unreliable performance. This outcome is expected, as the model does not support any built-in class balancing, making it poorly suited for highly imbalanced data.

The Random Forest classifier achieved accuracy comparable to the dummy reference, while its F1-score remained modest (around 14%). This is in line with expectations: although the model supports class balancing, it can still be biased toward the majority class due to the nature of majority voting in ensemble predictions.

Both Logistic Regression and Support Vector Machine (SVM) achieved the best performance in terms of F1-score (approximately 20–30%) and showed accuracy clearly between the dummy classifier baselines. These results are consistent with the models’ ability to incorporate class weighting, which helps mitigate the impact of class imbalance.

Below, we present detailed results for lysine (the most imbalanced class) and proline (which shows a class imbalance level comparable to the remaining amino acids). Full results for the missing amino acids are provided in the **Supplementary Data [7.2]**.

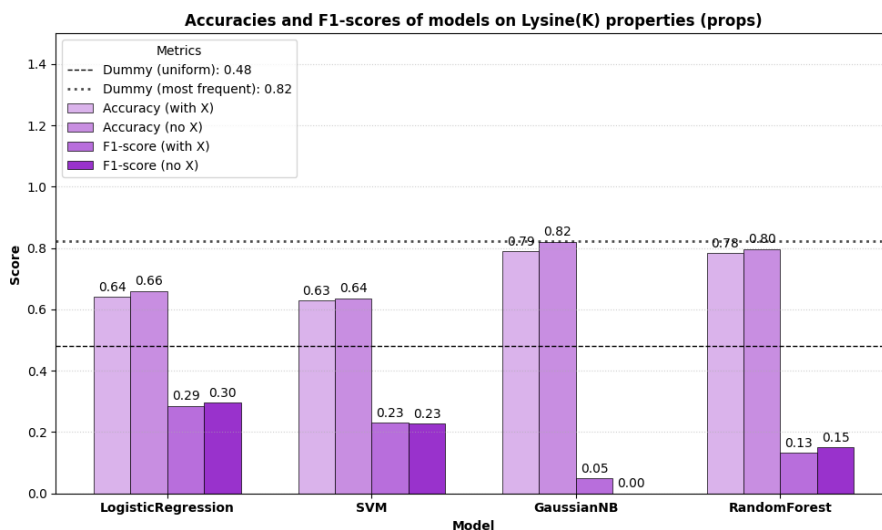


Figure 2: Lysine(K) - features based on sequence

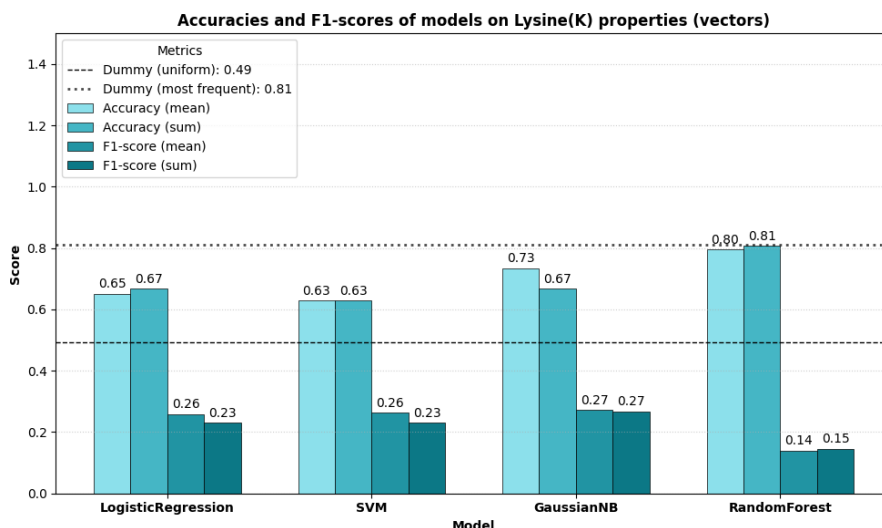


Figure 3: Lysine(K) - features based on amino acids

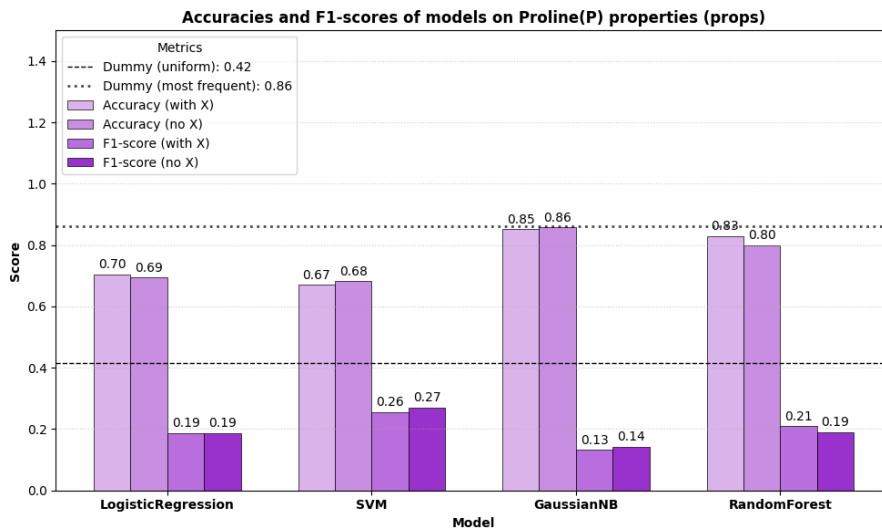


Figure 4: Proline(P) - features based on sequence

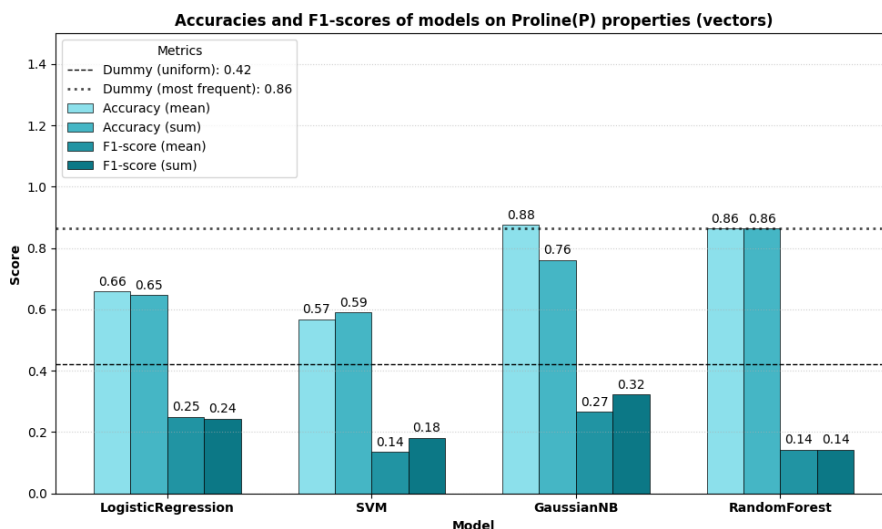


Figure 5: Proline(P) - features based on amino acids

## 5 Deep learning models - LSTM

### 5.1 Description

The model is a neural network used to classify sequences into multiple classes. It consists of one LSTM layer that learns patterns over time, followed by a dropout layer to reduce overfitting, and a dense output layer with softmax activation to generate class probabilities. The model is trained using the *Adam* optimizer and a loss function that compares predicted probabilities with the correct class (given as an integer). Additionally, class weights are calculated and applied during training to compensate for imbalanced data, ensuring that underrepresented classes have a stronger influence on the learning process.

## 5.2 Results

The LSTM model demonstrates a gradual improvement in training accuracy across epochs, stabilizing around 50–60% on the training dataset. Training loss consistently decreases, suggesting ongoing optimization. Final F1-scores between 0.25 and 0.28 indicate moderate sensitivity to the minority class, which is a favorable result given the high class imbalance. Overall, the training dynamics suggest stable learning with limited overfitting.

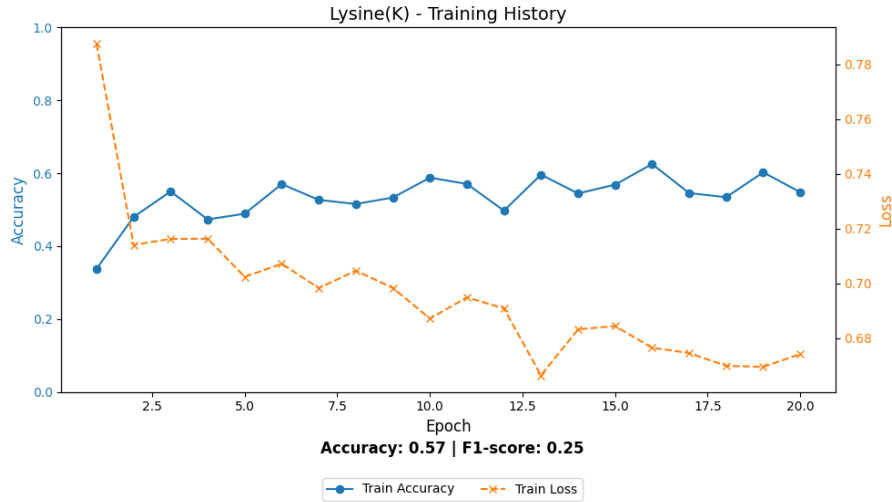


Figure 6: Lysine(K) - LSTM

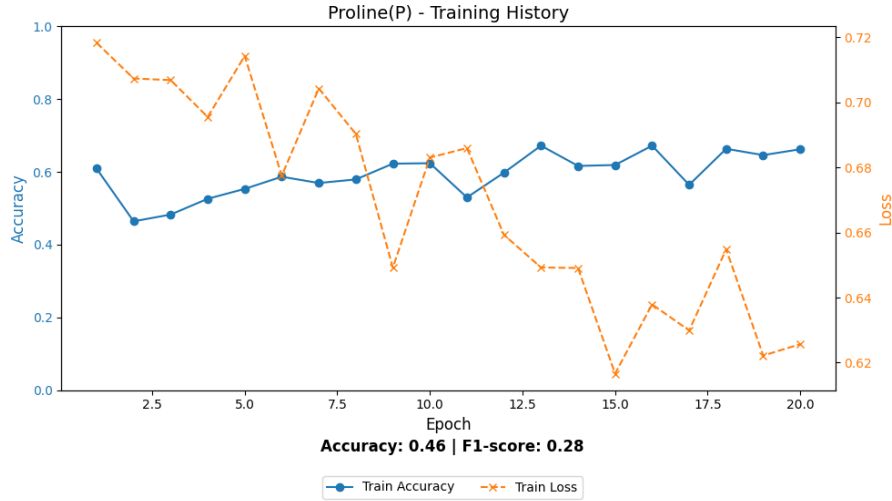


Figure 7: Proline(P) - LSTM

## 6 Conclusion

The evaluation of traditional machine learning models across various input representations reveals a notable trade-off between overall accuracy and class-specific performance as measured by the F1-score.

|                 | Models       | Logistic Regression | SVM  | Gaussian NB | RandomForest | Mean  | LSTM |
|-----------------|--------------|---------------------|------|-------------|--------------|-------|------|
| <b>Accuracy</b> | props        | 0.7                 | 0.67 | 0.85        | 0.83         | 0.763 | n.a  |
|                 | props no x   | 0.69                | 0.68 | 0.86        | 0.8          | 0.758 | n.a  |
|                 | mean vectors | 0.66                | 0.57 | 0.88        | 0.86         | 0.743 | n.a  |
|                 | sum vectors  | 0.65                | 0.59 | 0.76        | 0.86         | 0.715 | n.a  |
|                 | embeddings   | n.a                 | n.a  | n.a         | n.a          | n.a   | 0.48 |
| <b>F1-score</b> | props        | 0.19                | 0.26 | 0.13        | 0.21         | 0.198 | n.a  |
|                 | props no x   | 0.19                | 0.27 | 0.14        | 0.19         | 0.198 | n.a  |
|                 | mean vectors | 0.25                | 0.14 | 0.27        | 0.14         | 0.2   | n.a  |
|                 | sum vectors  | 0.24                | 0.18 | 0.32        | 0.14         | 0.22  | n.a  |
|                 | embeddings   | n.a                 | n.a  | n.a         | n.a          | n.a   | 0.29 |

Table 1: Performance of traditional machine learning models and LSTM across various input representations. Accuracy and F1-score are reported for each model-input combination. LSTM was only evaluated on sequence embeddings.

While Gaussian Naive Bayes consistently achieves the highest accuracy (up to 88%), its low F1-score (0.13–0.32) suggests poor performance on the minority class, likely due to the lack of built-in class balancing. Logistic Regression and SVM, both with class weighting enabled, show a more balanced trade-off, with F1-scores reaching up to 0.27 and overall accuracy remaining within an acceptable range (0.65–0.70). Random Forest, despite its class balancing support, exhibits high accuracy (up to 0.86) but relatively low F1-scores (0.14–0.21), likely due to bias from the majority voting mechanism.

The LSTM model—trained directly on sequence embeddings—achieves the highest F1-score (0.29), despite lower overall accuracy (0.48). This suggests that LSTMs, while less effective in modeling the majority class globally, are more capable of capturing sequence-level patterns relevant to the minority class, making them better suited for highly imbalanced classification tasks.

In summary, when addressing strong class imbalance, models with explicit class weighting (SVM, Logistic Regression) or sequence modeling capabilities (LSTM) provide more reliable detection of the minority class compared to models that prioritize global accuracy.



## 7 Supplementary Data

### 7.1 Data indenpence

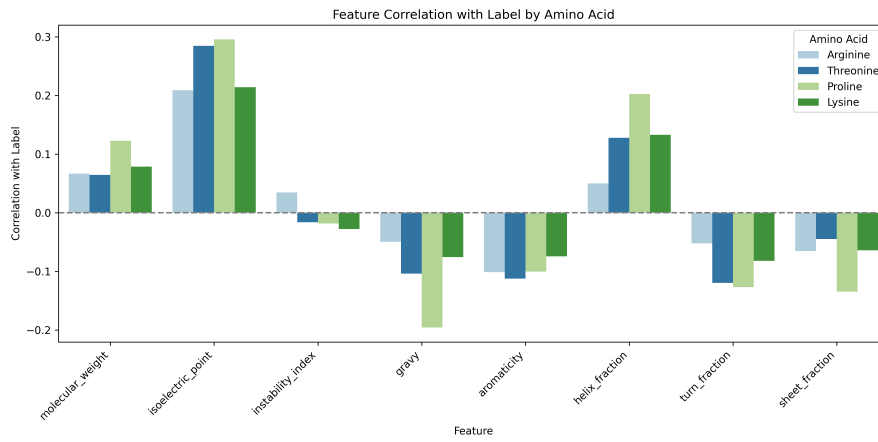


Figure 8: Weak correlation between the label and features based on sequence

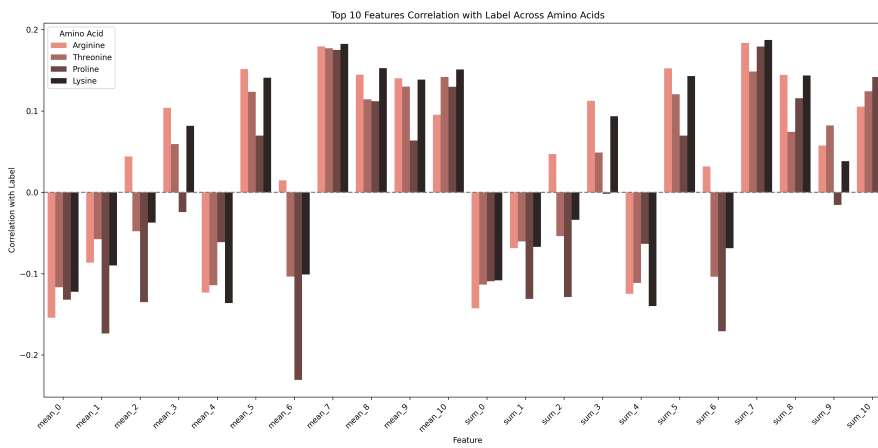


Figure 9: Weak correlation between the label and features based on amino acids

## 7.2 Graphs for remaining amino acids

### Classical Supervised Learning Models

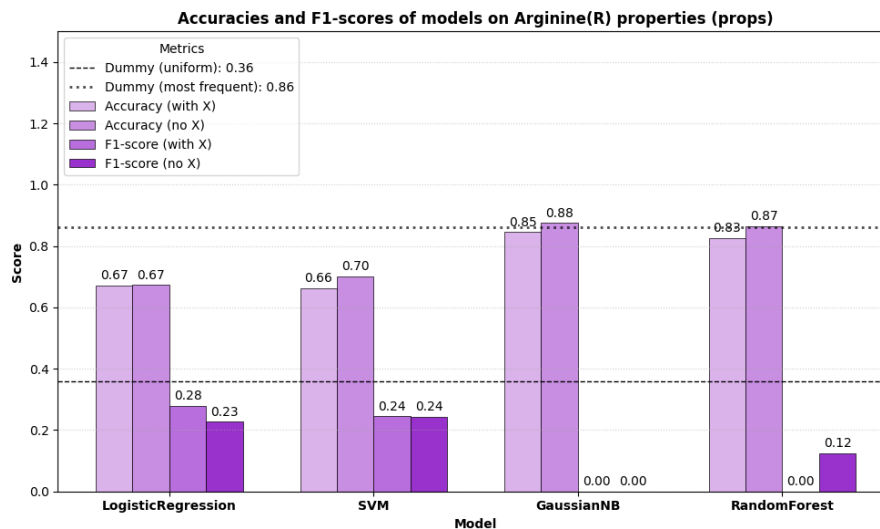


Figure 10: Arginine(R) - features based on sequence

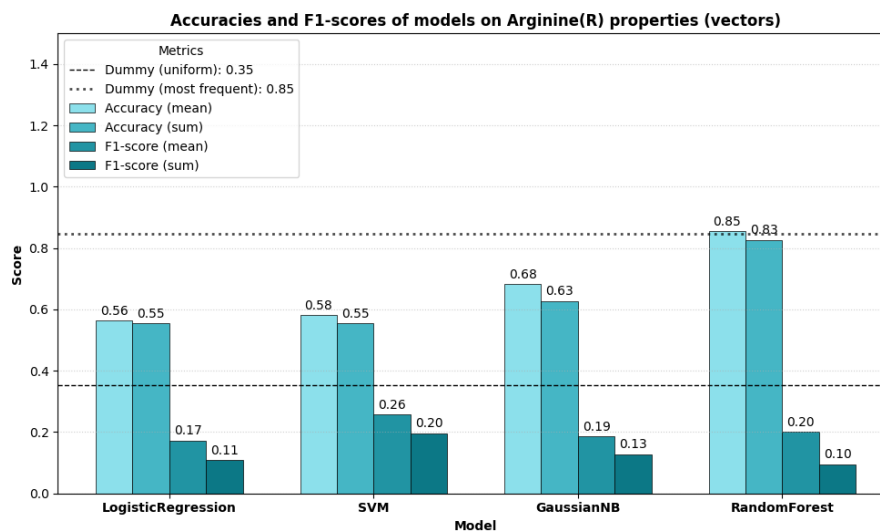


Figure 11: Arginine(R) - features based on amino acids

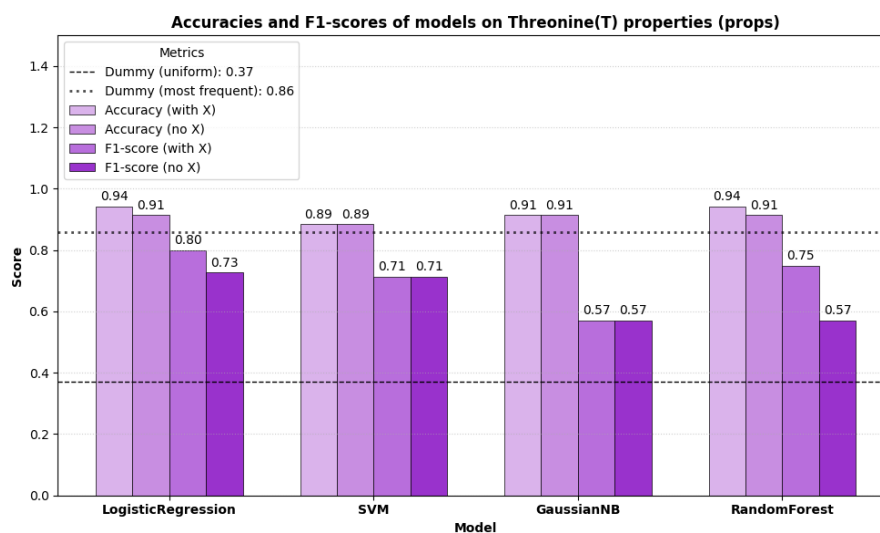


Figure 12: Threonine(T) - features based on sequence

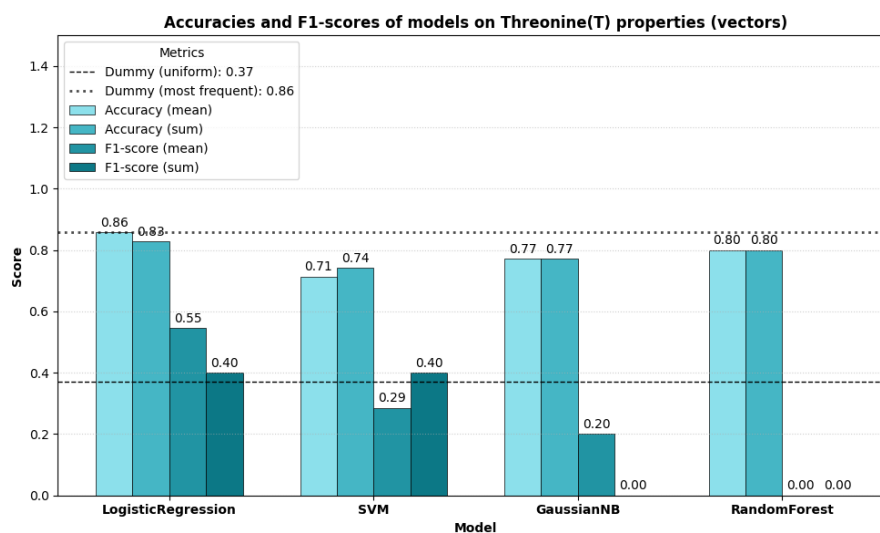


Figure 13: Threonine(T) - features based on amino acids

# LSTM

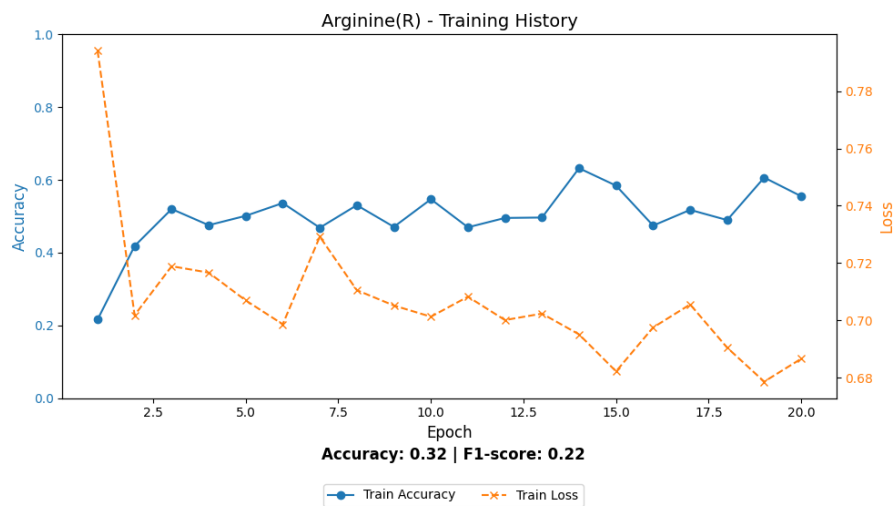


Figure 14: Arginine (R) - LSTM

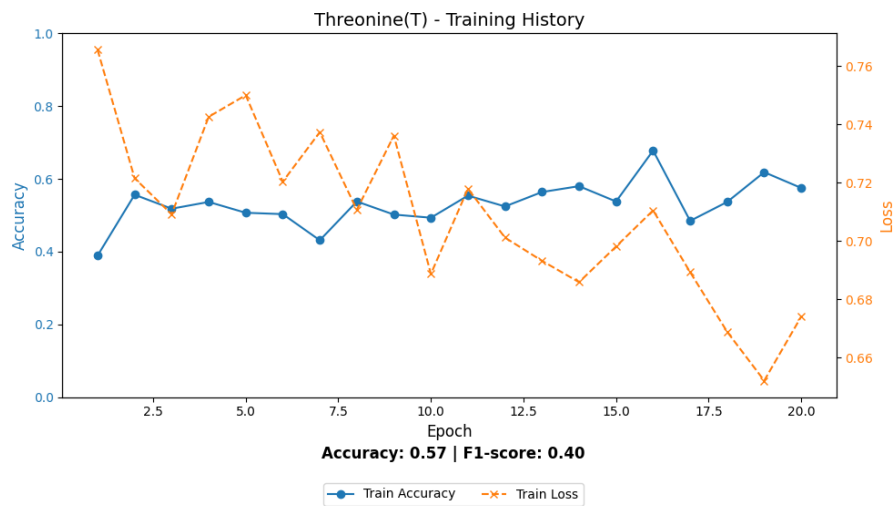


Figure 15: Threonine (T) - LSTM

## 8 Bibliography

1. Dan Zhang, Zhao-Chun Xu, Wei Su, Yu-He Yang, Hao Lv, Hui Yang, Hao Lin, iCarPS: a computational tool for identifying protein carbonylation sites by novel encoded features, Bioinformatics, Volume 37, Issue 2, January 2021, Pages 171–177, <https://doi.org/10.1093/bioinformatics/btaa702>